The (failure of the) variational principle for slow entropy

Nicanor Carrasco-Vargas

Jagiellonian University in Krakow

Conventions

Interrupt me as much as possible.

Talk summary

- I will speak about dynamical systems :)
- People study something called *entropy* ✓
- Entropy satisfies a nice property called *variational principle* ✓.
- There are several entropy-type objects, one of them being *slow entropy*.
- It sometimes fails the variational principle : (

Dynamical system $\approx$ some bijection of a set

$$T : X \to X$$

that preserves some extra mathematical structure.

Topological dynamical system

A homeomorphism of a compact metrizable topological space.

Measure-preserving dynamical system

An automorphism of a probability space.

Notation

Topological dynamical system = (X, T)

Measure-preserving system = $(X, \mathcal{B}, \mu, T)$

Topological $\to$ measure-preserving

How to get a measure-preserving system from a topological one?

Let (X, T) be a topological dynamical system.
$\mathcal{B}$ = Borel sigma-algebra on X.
μ = Borel probability measure on X which is T-**invariant**:

$$\mu(A) = \mu(T^{-1}(A)),\ A \in \mathcal{B}.$$

Then $(X, \mathcal{B}, \mu)$ is a probability space and T is an automorphism.

Topological $\rightarrow$ measure-preserving

Definition

Let $\mathcal{M}(T)$ be the set of Borel probability measures on X that are T-invariant.

Each $\mu \in \mathcal{M}(T)$ gives us a new measure-preserving system $(X, \mathcal{B}, \mu, T)$.

The interplay of topology and measures

Topological $\leftrightarrow$ measure-preserving dynamic

A beautiful theorem about this interplay is the **variational principle** for entropy.

What is entropy?

Entropy measures the abundance of something. What is this something?

- Chaos
- Disorder
- Information
- Distinguishable orbit segments
- Bits of information
- (uncountably many interpretations)

Topological entropy

The topological entropy of (X, T) is

$$h_{top}(T) := \sup_{\mathcal{U}} \lim_{n\to\infty} \frac{1}{n} \log(N(\mathcal{U}_0^{n-1}))$$

Where

- $\mathcal{U}$ ranges over finite open covers of X.
- $N(\mathcal{U})$ denotes the minimal cardinality of a subcover of $\mathcal{U}$.
- $\mathcal{U}_0^{n-1}$ denotes $\vee_{i=0}^{n-1} T^{-i}(\mathcal{U})$
  - $T^{-i}(\mathcal{U}) = \{T^{-i}(U) : U \in \mathcal{U}\}$
  - $\mathcal{U} \vee \mathcal{V} = \{U \cap V : U \in \mathcal{U},\ V \in \mathcal{V}\}$

Measure-theoretic entropy

The measure-theoretic entropy of $(X, \mathcal{B}, \mu, T)$ is

$$h_\mu(T) = \sup_{\mathcal{P}} \lim_{n\to\infty} \frac{1}{n} H_\mu(\mathcal{P}_0^{n-1})$$

Where

- $\mathcal{P}$ ranges over finite measurable partitions of X.
- $H_\mu(\mathcal{P}) = -\sum_{P\in\mathcal{P}} \mu(P) \log \mu(P)$
- $\mathcal{P}_0^{n-1}$ denotes $\vee_{i=0}^{n-1} T^{-i}(\mathcal{P})$
  - $T^{-i}(\mathcal{P}) = \{T^{-i}(P) : P \in \mathcal{P}\}$
  - $\mathcal{P} \vee \mathcal{Q} = \{P \cap Q : P \in \mathcal{P},\ Q \in \mathcal{Q}\}$

Topological $\Longleftrightarrow$ measure-preserving dynamics

Theorem (the variational principle)

For a topological dynamical system (X, T) we have

$$\sup_{\mu \in \mathcal{M}(T)} h_\mu(T) = h_{top}(T)$$

Questions people study:

- Is there a measure realizing this supremum?
- Is there only one?
- ...

Zero entropy systems

How to distinguish two zero entropy systems?

Slow entropy

Entropy captures the exponential growth rate of something. For instance, $h_{top}(T) = 3$ when

$$N(\mathcal{U}_0^{n-1}) \approx e^{3n}$$

Our mission is to capture sub-exponential growth rates. We can do this with slow entropy.

Scales

Definition

A **scale a** $= \{a_n(t)\}_{n\in\mathbb{N},t>0}$ is a family of functions such that

- $a_n(t) \in (0, \infty)$
- $a_n(t)$ non-decreasing in t
- $\lim_{n\to\infty} a_n(t) = \infty$ for $t > 0$

Some scales

- The exponential scale $exp = \{e^{nt}\}_{n\in\mathbb{N},t>0}$
- The polynomial scale $pol = \{n^t\}_{n\in\mathbb{N},t>0}$
- The stretched exponential scale $\{e^{n^t}\}_{n\in\mathbb{N},t>0}$

Topological slow entropy

The *upper* topological slow entropy of (X, T) with scale $\mathbf{a} = \{a_n(t)\}_{n\in\mathbb{N},t>0}$ is

$$\overline{ent}^{\mathbf{a}}_{top}(T) = \sup_{\mathcal{U}} \sup(\{0\} \cup \{t > 0 : \limsup_{n\to\infty} \frac{N(\mathcal{U}_0^{n-1})}{a_n(t)} > 0\})$$

One similarly defines lower slow entropy $\underline{ent}^{\mathbf{a}}_{top}(T)$, with a liminf.

Measure-theoretic slow entropy

The *upper* measure-theoretic slow entropy of $(X, \mathcal{B}, \mu, T)$ with scale $\mathbf{a} = \{a_n(t)\}_{n\in\mathbb{N},t>0}$ is

$$\overline{ent}^{\mathbf{a}}_{\mu}(T) = \sup_{\mathcal{P}} \sup_{\epsilon,\delta>0} \sup(\{0\} \cup \{t > 0 : \limsup_{n\to\infty} \frac{S^H_{\mathcal{P}}(n, \epsilon, \delta)}{a_n(t)} > 0\})$$

Here $S^H_{\mathcal{P}}(n, \epsilon, \delta)$ denotes the minimal number of ϵ-balls in $d^H_{\mathcal{P},n}$ needed to cover measure $1 - \delta$. Here

$$d^H_{\mathcal{P},n}(x, y)$$

is the proportion of "times" $i \in \{0, \ldots, n-1\}$ such that $T^i(x)$ and $T^i(y)$ fall into different atoms of $\mathcal{P}$.
One similarly defines lower slow entropy $\underline{ent}^{\mathbf{a}}_{\mu}(T)$, with liminf instead of limsup.

One inequality from the variational principle

Theorem

For a topological system (X, T) and a scale $\mathbf{a}$, we have

$$\sup_{\mu\in\mathcal{M}(T)} \overline{ent}^{\mathbf{a}}_{\mu}(T) \leq \overline{ent}^{\mathbf{a}}_{top}(T)$$

$$\sup_{\mu\in\mathcal{M}(T)} \underline{ent}^{\mathbf{a}}_{\mu}(T) \leq \underline{ent}^{\mathbf{a}}_{top}(T)$$

Question

When is it true that

$$\sup_{\mu \in \mathcal{M}(T)} ent^{\mathbf{a}}_{\mu}(T) = ent^{\mathbf{a}}_{top}(T)?$$

What do we know?

There are several positive and negative examples (Kanigowski, Katok, Wei, Vinhage, Kunde, Banerjee, ...)

There is much less structural understanding than with classical entropy.

Main result

Theorem (C, 2026)

For the polynomial scale $pol = \{n^t\}_{n\in\mathbb{N},t>0}$, there are examples of topological dynamical systems (X, T) with

$$\sup_{\mu\in\mathcal{M}(T)} \underline{ent}_{\mu}^{pol}(T) < \underline{ent}_{top}^{pol}(T)$$

such that

- One can prescribe the set of invariant measures $\mathcal{M}(T)$ (up to affine homeomorphism).
- No ergodic measure in $\mathcal{M}(T)$ is Kronecker / discrete spectrum / a rotation.

Thanks for your attention